Kanjanapong Singhatap (Arm)



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Professional Summary

Versatile **Mechanical Design Engineer with over 5 years** of experience in product development, aluminum extrusion die design, sheet metal fabrication, structural design, and mechanical systems. Proficient in CAD modeling, engineering simulation (FEA and CFD), product testing, and cross-functional collaboration from concept to production. Experienced in design and testing planning, department KPI management, data analysis, and monthly performance presentation to management. Demonstrated ability to coordinate engineering activities, lead design teams, optimize cost and processes, and solve real-world engineering problems through practical and data-driven solutions.

Core Competencies

- **Mechanical Design & Product Development:** Mechanical Component Design, Assembly Design, Product Development, Design for Manufacturing (DFM), Design for Assembly (DFA), Design Optimization
- **3D CAD & Engineering Design: SolidWorks, Autodesk Inventor, Fusion 360, AutoCAD**, 3D Modeling, Assembly Modeling, 2D Technical Drawing, Design Modification
- **Aluminum Extrusion Die Design:** Aluminum Extrusion Die Design, Profile Development, Die Design Optimization, Die Trial & Validation, QForm Flow Simulation
- **Sheet Metal & Fabrication Design:** Sheet Metal Design, Flat Pattern Development, Laser Cutting, Bending/Forming, Welding Design, Fabrication Drawing
- **Engineering Analysis & Simulation: FEA, CFD, QForm Simulation, Autodesk Nastran**, Structural Analysis, Design Validation, Engineering Data Analysis
- **Manufacturing & Production:** Manufacturing Process Understanding, CNC Machining, Laser Cutting, Forming, Welding, Aluminum Extrusion, Assembly, Inspection & Quality Control
- **Design for Manufacturing:** Manufacturability Evaluation, Material Selection, Tolerance Consideration, Production Constraints, Cost Optimization, Process Improvement
- **Testing & Validation:** Prototype Testing, Die Trial Planning, Product Testing, Test Result Analysis, Root Cause Analysis, Corrective & Preventive Actions
- **Technical Documentation:** Engineering Drawing, GD&T, BOM, Manufacturing Documentation, Assembly Documentation, Technical Handover
- **Engineering Planning & Coordination:** Design Planning, Test & Trial Planning, Project Scheduling, Progress Monitoring, Cross-functional Coordination
- **KPI & Technical Management:** Department KPI Monitoring, Data Collection & Analysis, Monthly KPI Reporting, Management Presentation, Continuous Improvement

Professional Experience

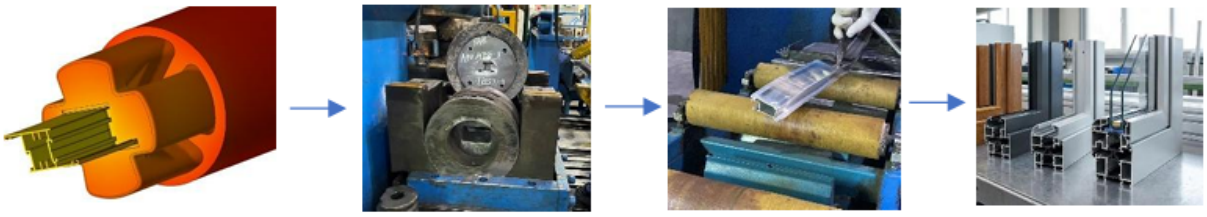
Alumet Co., Ltd.

Die Design Engineer / Acting Assistant Manager Responsibilities | *Oct 2025 – Present*

- Design aluminum extrusion dies for various profiles, including door/window frames, heatsinks, and custom sections.
- Perform aluminum extrusion flow simulation using QForm to evaluate die strength, metal flow balance, extrusion velocity, and exit temperature before production.
- Plan and manage the **die design process**, including design priorities, development schedules, technical reviews, and design completion targets.
- Plan and coordinate **die testing and extrusion trials**, including test objectives, simulation parameters, trial conditions, and evaluation criteria.
- Analyze trial and inspection results to identify design issues, determine root causes, and implement corrective actions to improve die performance and product quality.
- Collect, consolidate, and analyze **department performance data** for monthly KPI reporting.
- Prepare and present **monthly department KPI performance** to management, including performance results, issues, improvement actions, and progress against targets.
- Monitor department workload, project progress, testing status, and key performance indicators to support effective resource and schedule planning.
- Coordinate with production, quality, planning, and related departments to resolve technical issues and ensure projects are completed according to schedule.
- Prepare technical drawings and documentation and ensure all die design stages are completed on time.
- Record inspection and trial results and support sample qualification and delivery to customers.
- Participate in troubleshooting and improvement meetings, providing engineering solutions based on simulation, testing, and production data.

Key Achievements:

- Designed new extrusion dies using QForm, SolidWorks, and AutoCAD with FEA validation to improve die performance and reduce trial runs.
- Successfully tested new products and delivered qualified samples to customers on schedule
- Successfully took responsibility for **department-level KPI data collection, analysis, and monthly management presentation**.
- Expanded responsibilities beyond individual die design to include **design planning, testing planning, performance monitoring, and department-level coordination**.
- Improved the connection between **simulation → die design → trial testing → data analysis → corrective action**, supporting more systematic product development.
- Successfully tested new extrusion dies and delivered qualified samples to customers according to schedule.



ANCA Manufacturing (Thailand) Co., Ltd.

Design Engineer | *Mar 2024 – Apr 2025*

- Reviewed customer-provided **2D drawings and 3D CAD models** upon receiving new product purchase orders, identifying design requirements, critical dimensions, and potential manufacturing issues.
- Evaluated **Design for Manufacturing (DFM)** to ensure customer designs were suitable for in-house production processes and manufacturing capabilities.
- Evaluated manufacturability with a focus on **sheet metal forming, welding, CNC machining, and fabrication processes**.
- Developed and optimized **3D CAD models and sheet metal flat patterns** for laser cutting and forming processes while considering material behavior, bend conditions, weldability, and assembly requirements.
- Designed and modified mechanical components to improve **manufacturability, assembly, dimensional accuracy, and production efficiency**.
- Prepared production-ready **2D technical drawings** with appropriate dimensions, tolerances, and GD&T requirements for manufacturing and inspection.
- Generated complete engineering documentation, including **MFEA reports, BOM, technical drawings, and manufacturing documentation**.
- Conducted design verification before releasing drawings to production to minimize design-related issues, rework, and manufacturing delays.
- Participated in **technical handover meetings** with Production, QC, Planning, Purchasing, Materials, and Safety teams to communicate design requirements and production considerations.
- Presented **production timelines and project requirements**, clarified design intent, and supported cross-functional teams throughout the product development process.
- Worked closely with the **Sales team and customers** to resolve drawing issues, clarify technical requirements, and address additional design requests.
- Followed up on **daily production progress**, monitored project status, and identified potential bottlenecks that could affect delivery schedules.
- Provided real-time **engineering support to Production and QC teams** when design, dimensional, fabrication, or assembly issues occurred during manufacturing.
- Investigated production problems and supported **root cause analysis and corrective actions** to resolve design and manufacturing issues.
- Coordinated with relevant departments to ensure engineering changes and technical requirements were properly communicated and implemented.
- Optimized designs based on **actual manufacturing capabilities**, production feedback, and fabrication requirements to improve overall product quality, cost, and production efficiency.
- Supported the complete **new product development process from customer drawing review through design, manufacturing, inspection, and final product delivery**.

Key Achievements:

- Reduced **design-related rework** through improved drawing review and design verification processes before production release.
- Successfully delivered multiple **new products according to the required production schedule**, with no production delays caused by design issues.

- Optimized product designs to better match **in-house fabrication capabilities**, improving manufacturability, production efficiency, cost, and time-to-market.
- Improved communication between **Engineering, Production, QC, Planning, Purchasing, Materials, and Sales** through structured technical handover and design clarification.
- Provided effective engineering support during production to identify and resolve issues quickly, helping maintain **product quality and delivery schedules**.






SolarD Company Limited

Senior Mechanical Design Engineer | *Jan 2023 – Feb 2024 (Solar-D)*

- **Led the redesign of rooftop and ground-mounted solar panel mounting systems** with a strong focus on structural integrity, ease of installation, and cost-effectiveness.
- Supervised a team of 2 junior engineers, assigning tasks, reviewing outputs, and providing technical guidance.
- Utilized Autodesk Inventor to create precise 3D CAD models, 2D drawings, and assembly documentation.
- Performed FEA analysis using Autodesk Nastran to verify structural performance under real-world loads, including wind and weight of solar panels.
- **Spearheaded a major cost-reduction project that cut total structure cost by up to 40%, achieved through part count minimization, material optimization, and fabrication simplification.**
- Sourced new component suppliers and negotiated with fabricators for prototype and mass production.
- Conducted real-world field installation tests and incorporated installer feedback into product revisions.
- **Delivered full production packages: BOM, drawings, specifications, and assembly instructions to the production team.**
- Led handover meetings and closely monitored manufacturing to address any production challenges.

Key Achievements:

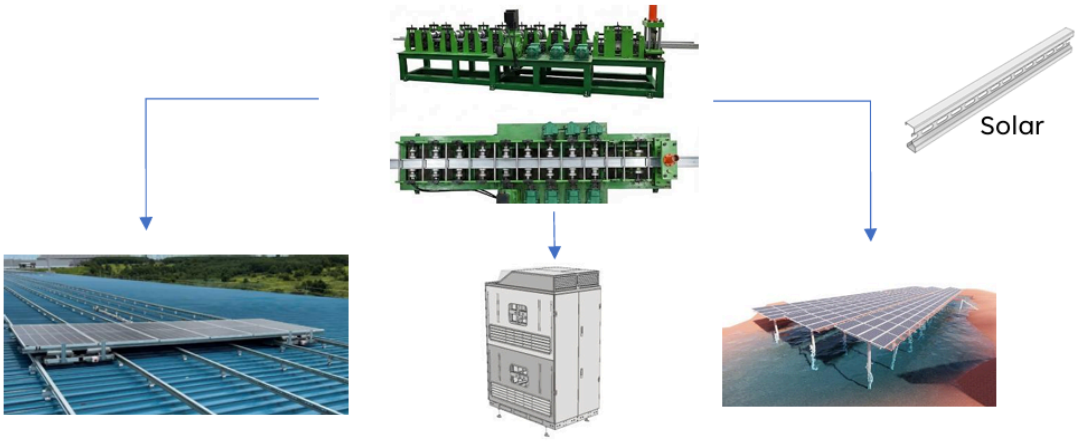
- **Reduced mounting system installation time by 25%.**
- Successfully launched a new mounting product line used in multiple commercial-scale projects.
- Developed internal documentation templates now used across the design department.

Mechanical Design Engineer | *Nov 2021 – Dec 2023 (Solar-D)*

- Designed solar mounting solutions tailored to various rooftop types (metal sheet, tile, concrete).
- Built a library of modular design templates for rapid reuse across multiple projects.
- Coordinated with the electrical team to ensure layout compatibility with inverter and cable routing.
- **Created and updated technical drawings, BOMs, and installation manuals for field engineers**.
- Supported procurement with material selection, cost comparison, and technical clarification.
- Assisted sales team with drawing presentations and client-specific layout proposals.

Key Achievements:

- Developed a standardized design process, reducing design turnaround time by 30%.
- Promoted to Senior Mechanical Design Engineer within a year due to consistent performance and technical leadership.



Education

B.Eng. in Automotive Engineering
Suranaree University of Technology | *2018 – 2021*